

```
mysql> /* Numeric Functions:
/*> 1. abs() = absolute value of a number
/*> 2. ceil() = round up
/*> 3. floor() = round down
/*> 4. round() = round to a specific number of digits
/*> 5. mod() = modulo of a number
/*> 6. pow()/power() = exponentiation
/*> 7. sqrt() = square root
/*> 8. exp() = exponential function
/*> 9. radians() = convert degrees to radians
/*> 10.degrees() = convert radians to degrees
/*> 11.sign() = returns the sign of a number
/*> 12.rand() = generates a random number
/*> 13.greatest() = find the maximum value
/*> 14.least() = find the minimum value
/*> 15.pi() = returns the value of pi
/*> 16.hex() = convert to hexadecimal
/*> 17.oct() = convert to octal
/*> 18.bin() = convert to binary
/*> */
```

```
mysql> use cg_mysql;
```

```
Database changed
```

```
mysql> select abs(-100) as absolute_value;
```

```
+-----+
| absolute_value |
+-----+
|          100 |
+-----+
```

```
1 row in set (0.01 sec)
```

```
mysql> select ceil(8.3) as ceil_value;
```

```
+-----+
| ceil_value |
+-----+
|          9 |
+-----+
```

```
1 row in set (0.04 sec)
```

```
mysql> select floor(8.3) as floor_value;
```

```
+-----+
| floor_value |
+-----+
|          8 |
+-----+
```

```
1 row in set (0.04 sec)
```

```
mysql> select round(8.264982,2) as rounded_value;
```

```
+-----+
| rounded_value |
+-----+
```

```
|          8.26 |
+-----+
1 row in set (0.06 sec)
```

```
mysql> select mod(45,3) as reminder;
```

```
+-----+
| reminder |
+-----+
|         0 |
+-----+
1 row in set (0.04 sec)
```

```
mysql> select pow(3,4) as power_value;
```

```
+-----+
| power_value |
+-----+
|          81 |
+-----+
1 row in set (0.00 sec)
```

```
mysql> select sqrt(45) as square_root;
```

```
+-----+
| square_root |
+-----+
| 6.708203932499369 |
+-----+
1 row in set (0.00 sec)
```

```
mysql> select round(sqrt(45),2) as rounded_square_root;
```

```
+-----+
| rounded_square_root |
+-----+
|          6.71 |
+-----+
1 row in set (0.00 sec)
```

```
mysql> select ln(10) as natural_log;
```

```
+-----+
| natural_log |
+-----+
| 2.302585092994046 |
+-----+
1 row in set (0.00 sec)
```

```
mysql> select log(10,100) as log_value;
```

```
+-----+
| log_value |
+-----+
|         2 |
+-----+
```

1 row in set (0.00 sec)

```
mysql> select pi() as pi_value;
```

pi_value
3.141593

1 row in set (0.03 sec)

```
mysql> select radians(30) as degrees_to_radians;
```

degrees_to_radians
0.5235987755982988

1 row in set (0.05 sec)

```
mysql> select degrees(0.523) as radians_to_degrees;
```

radians_to_degrees
29.965692685342056

1 row in set (0.03 sec)

```
mysql> select sign(-125) as sign_value;
```

sign_value
-1

1 row in set (0.05 sec)

```
mysql> select sign(125) as sign_value;
```

sign_value
1

1 row in set (0.00 sec)

```
mysql> select sign(0) as sign_value;
```

sign_value
0

1 row in set (0.00 sec)

```
mysql> select rand() as random_number;
```

```
+-----+
| random_number |
+-----+
| 0.8549883755436058 |
+-----+
1 row in set (0.06 sec)
```

```
mysql> select rand() as random_number;
```

```
+-----+
| random_number |
+-----+
| 0.9087207754223801 |
+-----+
1 row in set (0.00 sec)
```

```
mysql> select rand(1000,9999) as random_number;
```

```
ERROR 1582 (42000): Incorrect parameter count in the call to native function 'rand'
```

```
mysql> select randint(100,400) as random_number;
```

```
ERROR 1305 (42000): FUNCTION cg_mysql.randint does not exist
```

```
mysql> select greatest(10,-20,35,70) as greatest_value;
```

```
+-----+
| greatest_value |
+-----+
|          70 |
+-----+
1 row in set (0.00 sec)
```

```
mysql> select least(10,-20,35,70) as greatest_value;
```

```
+-----+
| greatest_value |
+-----+
|          -20 |
+-----+
1 row in set (0.00 sec)
```

```
mysql> select bit_count(10) as bit_count;
```

```
+-----+
| bit_count |
+-----+
|          2 |
+-----+
1 row in set (0.03 sec)
```

```
mysql> select bit_count(7) as bit_count;
```

```
+-----+
| bit_count |
+-----+
|          3 |
+-----+
```

1 row in set (0.00 sec)

mysql> -- as 10 = 1010 and 7 = 0111 bit_count() gives number of 1s in a digit after converting into binary;

mysql> select bin(15) as decimal_value;

decimal_value
1111

1 row in set (0.02 sec)

mysql> select oct(15) as octal_value;

octal_value
17

1 row in set (0.00 sec)

mysql> select hex(15) as hexadecimal_value;

hexadecimal_value
F

1 row in set (0.05 sec)

mysql> show tables;

Tables_in_cg_mysql
backup_emp
company_staff
departments
doctors
emp
employees
orders
products
projects
sales
staff
students
studentscores

13 rows in set (0.06 sec)

mysql> drop table employees;

Query OK, 0 rows affected (0.07 sec)

```
mysql> CREATE TABLE employees (  
-> emp_id INT PRIMARY KEY,  
-> first_name VARCHAR(30),  
-> last_name VARCHAR(30),  
-> email VARCHAR(60),  
-> phone_number VARCHAR(20),  
-> department VARCHAR(30),  
-> city VARCHAR(30),  
-> salary DECIMAL(10,2)  
-> );  
Query OK, 0 rows affected (0.05 sec)
```

```
mysql>  
mysql>  
mysql> INSERT INTO employees VALUES  
->  
(101,'Rahul','Kumar','rahul.kumar@gmail.com','987-654-3210','IT','Hyderabad',45000)  
,  
->  
(102,'Anita','Sharma','anita.sharma@yahoo.com','912-345-6789','HR','Bangalore',3800  
0),  
->  
(103,'Suresh','Verma','suresh.verma@gmail.com','998-123-4567','Sales','Chennai',520  
00),  
->  
(104,'Priya','Singh','priya.singh@outlook.com','900-456-7890','Finance','Pune',4700  
0),  
->  
(105,'Rahul','Joshi','rahul.joshi@gmail.com','955-888-1111','Testing','Delhi',41000  
),  
->  
(106,'Kiran','Reddy','kiran.reddy@gmail.com','901-222-3333','IT','Hyderabad',56000)  
,  
->  
(107,'Deepika','Naidu','deepika.naidu@yahoo.com','944-111-2222','HR','Vizag',39000)  
,  
->  
(108,'Mahesh','Patel','mahesh.patel@gmail.com','933-444-5555','Sales','Mumbai',6200  
0),  
->  
(109,'Divya','Gupta','divya.gupta@gmail.com','955-777-8888','Finance','Delhi',49000  
),  
->  
(110,'Arjun','Rao','arjun.rao@gmail.com','988-999-7777','Developer','Hyderabad',680  
00);  
Query OK, 10 rows affected (0.02 sec)  
Records: 10 Duplicates: 0 Warnings: 0
```

```
mysql> select * from employees;
```

emp_id	first_name	last_name	email	phone_number	department	city	salary
101	Rahul	Kumar	rahul.kumar@gmail.com	987-654-3210	IT	Hyderabad	45000.00
102	Anita	Sharma	anita.sharma@yahoo.com	912-345-6789	HR	Bangalore	38000.00
103	Suresh	Verma	suresh.verma@gmail.com	998-123-4567	Sales	Chennai	52000.00
104	Priya	Singh	priya.singh@outlook.com	900-456-7890	Finance	Pune	47000.00
105	Rahul	Joshi	rahul.joshi@gmail.com	955-888-1111	Testing	Delhi	41000.00
106	Kiran	Reddy	kiran.reddy@gmail.com	901-222-3333	IT	Hyderabad	56000.00
107	Deepika	Naidu	deepika.naidu@yahoo.com	944-111-2222	HR	Vizag	39000.00
108	Mahesh	Patel	mahesh.patel@gmail.com	933-444-5555	Sales	Mumbai	62000.00
109	Divya	Gupta	divya.gupta@gmail.com	955-777-8888	Finance	Delhi	49000.00
110	Arjun	Rao	arjun.rao@gmail.com	988-999-7777	Developer	Hyderabad	68000.00

10 rows in set (0.00 sec)

```
mysql> select concat(first_name,'',last_name) as full_name from employees;
```

full_name
RahulKumar
AnitaSharma
SureshVerma
PriyaSingh
RahulJoshi
KiranReddy
DeepikaNaidu
MaheshPatel
DivyaGupta
ArjunRao

10 rows in set (0.03 sec)

```
mysql> select concat(first_name,' ',last_name) as full_name from employees;
```

full_name

Rahul Kumar
Anita Sharma
Suresh Verma
Priya Singh
Rahul Joshi
Kiran Reddy
Deepika Naidu
Mahesh Patel
Divya Gupta
Arjun Rao

10 rows in set (0.00 sec)

```
mysql> select first_name, upper(first_name) as upper_name from employees;
```

first_name	upper_name
Rahul	RAHUL
Anita	ANITA
Suresh	SURESH
Priya	PRIYA
Rahul	RAHUL
Kiran	KIRAN
Deepika	DEEPIKA
Mahesh	MAHESH
Divya	DIVYA
Arjun	ARJUN

10 rows in set (0.00 sec)

```
mysql> select last_name, lower(last_name) as lower_name from employees;
```

last_name	lower_name
Kumar	kumar
Sharma	sharma
Verma	verma
Singh	singh
Joshi	joshi
Reddy	reddy
Naidu	naidu
Patel	patel
Gupta	gupta
Rao	rao

10 rows in set (0.00 sec)

```
mysql> select email,substring(email,1,10) as email_prefix from employees;
```

email	email_prefix
rahul.kumar@gmail.com	rahul.kuma
anita.sharma@yahoo.com	anita.shar
suresh.verma@gmail.com	suresh.ver
priya.singh@outlook.com	priya.sing
rahul.joshi@gmail.com	rahul.josh
kiran.reddy@gmail.com	kiran.redd
deepika.naidu@yahoo.com	deepika.na
mahesh.patel@gmail.com	mahesh.pat
divya.gupta@gmail.com	divya.gupt
arjun.rao@gmail.com	arjun.rao@

10 rows in set (0.00 sec)

```
mysql> select length(first_name) as first_name_length from employees;
```

first_name_length
5
5
6
5
5
5
7
6
5
5

10 rows in set (0.00 sec)

```
mysql> select first_name, length(first_name) as first_name_length from employees;
```

first_name	first_name_length
Rahul	5
Anita	5
Suresh	6
Priya	5
Rahul	5
Kiran	5
Deepika	7
Mahesh	6
Divya	5
Arjun	5

10 rows in set (0.00 sec)

```
mysql> select first_name, last_name, concat(first_name," ",last_name) as full_name,
```

```
length(full_name) as full_name_length from employees;
ERROR 1054 (42S22): Unknown column 'full_name' in 'field list'
mysql> select phone_number, replace(phone_number, '-', '') as clean_number from
employees;
```

phone_number	clean_number
987-654-3210	9876543210
912-345-6789	9123456789
998-123-4567	9981234567
900-456-7890	9004567890
955-888-1111	9558881111
901-222-3333	9012223333
944-111-2222	9441112222
933-444-5555	9334445555
955-777-8888	9557778888
988-999-7777	9889997777

10 rows in set (0.05 sec)

```
mysql> -- trim() function is used to remove spaces at the beginning and end of a
string similar to strip() function from python.
mysql> -- syntax: select trim(column_name) from table_name;
mysql> select concat_ws(',', first_name, last_name, email) as details from employees;
```

details
Rahul,Kumar,rahul.kumar@gmail.com
Anita,Sharma,anita.sharma@yahoo.com
Suresh,Verma,suresh.verma@gmail.com
Priya,Singh,priya.singh@outlook.com
Rahul,Joshi,rahul.joshi@gmail.com
Kiran,Reddy,kiran.reddy@gmail.com
Deepika,Naidu,deepika.naidu@yahoo.com
Mahesh,Patel,mahesh.patel@gmail.com
Divya,Gupta,divya.gupta@gmail.com
Arjun,Rao,arjun.rao@gmail.com

10 rows in set (0.00 sec)

```
mysql> select first_name, position('a' in first_name) as position_of_a from
employees;
```

first_name	position_of_a
Rahul	2
Anita	1
Suresh	0
Priya	5
Rahul	2

Kiran	4
Deepika	7
Mahesh	2
Divya	5
Arjun	1

+-----+

10 rows in set (0.00 sec)

```
mysql> select first_name, left(first_name,3) as left_short_name from employees;
```

first_name	left_short_name
Rahul	Rah
Anita	Ani
Suresh	Sur
Priya	Pri
Rahul	Rah
Kiran	Kir
Deepika	Dee
Mahesh	Mah
Divya	Div
Arjun	Arj

+-----+

10 rows in set (0.05 sec)

```
mysql> select first_name, right(first_name,3) as right_short_name from employees;
```

first_name	right_short_name
Rahul	hul
Anita	ita
Suresh	esh
Priya	iya
Rahul	hul
Kiran	ran
Deepika	ika
Mahesh	esh
Divya	vya
Arjun	jun

+-----+

10 rows in set (0.00 sec)

```
mysql> select last_name, reverst(last_name) as reversed_last_name from employees;
ERROR 1305 (42000): FUNCTION cg_mysql.reverst does not exist
```

```
mysql> select last_name, reverse(last_name) as reversed_last_name from employees;
```

last_name	reversed_last_name
Kumar	ramuK
Sharma	amrahS

Verma	amreV
Singh	hgniS
Joshi	ihsoJ
Reddy	yddeR
Naidu	udiaN
Patel	letaP
Gupta	atpuG
Rao	oaR

10 rows in set (0.00 sec)

```
mysql> select first_name, lpad(first_name,10,'*') as padded_name from employees;
```

first_name	padded_name
Rahul	*****Rahul
Anita	*****Anita
Suresh	****Suresh
Priya	*****Priya
Rahul	*****Rahul
Kiran	*****Kiran
Deepika	***Deepika
Mahesh	*****Mahesh
Divya	*****Divya
Arjun	*****Arjun

10 rows in set (0.04 sec)

```
mysql> select first_name, rpad(first_name,10,'*') as padded_name from employees;
```

first_name	padded_name
Rahul	Rahul*****
Anita	Anita*****
Suresh	Suresh*****
Priya	Priya*****
Rahul	Rahul*****
Kiran	Kiran*****
Deepika	Deepika***
Mahesh	Mahesh*****
Divya	Divya*****
Arjun	Arjun*****

10 rows in set (0.00 sec)

```
mysql> select first_name, rpad(first_name,7,'*') as padded_name from employees;
```

first_name	padded_name
Rahul	Rahul**

Anita	Anita**
Suresh	Suresh*
Priya	Priya**
Rahul	Rahul**
Kiran	Kiran**
Deepika	Deepika
Mahesh	Mahesh*
Divya	Divya**
Arjun	Arjun**

10 rows in set (0.00 sec)

```
mysql> select first_name, lpad(first_name,7,' ') as padded_name from employees;
```

first_name	padded_name
Rahul	Rahul
Anita	Anita
Suresh	Suresh
Priya	Priya
Rahul	Rahul
Kiran	Kiran
Deepika	Deepika
Mahesh	Mahesh
Divya	Divya
Arjun	Arjun

10 rows in set (0.00 sec)

```
mysql> select first_name, lpad(first_name,7,'-') as padded_name from employees;
```

first_name	padded_name
Rahul	--Rahul
Anita	--Anita
Suresh	-Suresh
Priya	--Priya
Rahul	--Rahul
Kiran	--Kiran
Deepika	Deepika
Mahesh	-Mahesh
Divya	--Divya
Arjun	--Arjun

10 rows in set (0.00 sec)

```
mysql> select first_name, ascii(first_name) as ascii_value from employees;
```

first_name	ascii_value
------------	-------------

Rahul	82
Anita	65
Suresh	83
Priya	80
Rahul	82
Kiran	75
Deepika	68
Mahesh	77
Divya	68
Arjun	65

+-----+

10 rows in set (0.00 sec)

mysql> -- it generates the ascii value of the first character in the string.

mysql> select * from employees where position('r' in first_name) = 1;

+-----+

emp_id	first_name	last_name	email	phone_number	
department	city	salary			

+-----+

101	Rahul	Kumar	rahul.kumar@gmail.com	987-654-3210	IT
	Hyderabad	45000.00			
105	Rahul	Joshi	rahul.joshi@gmail.com	955-888-1111	Testing
	Delhi	41000.00			

+-----+

+-----+

2 rows in set (0.00 sec)

mysql> select * from employees where position('yahoo' in email) > 0;

+-----+

emp_id	first_name	last_name	email	phone_number	
department	city	salary			

+-----+

102	Anita	Sharma	anita.sharma@yahoo.com	912-345-6789	HR
	Bangalore	38000.00			
107	Deepika	Naidu	deepika.naidu@yahoo.com	944-111-2222	HR
	Vizag	39000.00			

+-----+

+-----+

2 rows in set (0.00 sec)

mysql> select concat(first_name, ' ', last_name, ' ', 'from ', department) as details from employees;

+-----+

details

+-----+

Rahul Kumar from IT

+-----+

Anita Sharma from HR
Suresh Verma from Sales
Priya Singh from Finance
Rahul Joshi from Testing
Kiran Reddy from IT
Deepika Naidu from HR
Mahesh Patel from Sales
Divya Gupta from Finance
Arjun Rao from Developer

10 rows in set (0.00 sec)

```
mysql> select email,length(email) as email_length from employees;
```

email	email_length
rahul.kumar@gmail.com	21
anita.sharma@yahoo.com	22
suresh.verma@gmail.com	22
priya.singh@outlook.com	23
rahul.joshi@gmail.com	21
kiran.reddy@gmail.com	21
deepika.naidu@yahoo.com	23
mahesh.patel@gmail.com	22
divya.gupta@gmail.com	21
arjun.rao@gmail.com	19

10 rows in set (0.00 sec)

```
mysql> select * from employees where length(first_name) > 5;
```

emp_id	first_name	last_name	email	phone_number	department	city	salary
103	Suresh	Verma	suresh.verma@gmail.com	998-123-4567	Sales	Chennai	52000.00
107	Deepika	Naidu	deepika.naidu@yahoo.com	944-111-2222	HR	Vizag	39000.00
108	Mahesh	Patel	mahesh.patel@gmail.com	933-444-5555	Sales	Mumbai	62000.00

3 rows in set (0.00 sec)

```
mysql> select email, replace(email,'gmail.com','company.com') as email_replaced from employees;
```

email	email_replaced
-------	----------------

rahul.kumar@gmail.com	rahul.kumar@company.com
anita.sharma@yahoo.com	anita.sharma@yahoo.com
suresh.verma@gmail.com	suresh.verma@company.com
priya.singh@outlook.com	priya.singh@outlook.com
rahul.joshi@gmail.com	rahul.joshi@company.com
kiran.reddy@gmail.com	kiran.reddy@company.com
deepika.naidu@yahoo.com	deepika.naidu@yahoo.com
mahesh.patel@gmail.com	mahesh.patel@company.com
divya.gupta@gmail.com	divya.gupta@company.com
arjun.rao@gmail.com	arjun.rao@company.com

10 rows in set (0.00 sec)

```
mysql> select email, position('@' in email) as @_position from employees;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that
corresponds to your MySQL server version for the right syntax to use near
'@_position from employees' at line 1
```

```
mysql> select email, position('@' in email) from employees;
```

email	position('@' in email)
rahul.kumar@gmail.com	12
anita.sharma@yahoo.com	13
suresh.verma@gmail.com	13
priya.singh@outlook.com	12
rahul.joshi@gmail.com	12
kiran.reddy@gmail.com	12
deepika.naidu@yahoo.com	14
mahesh.patel@gmail.com	13
divya.gupta@gmail.com	12
arjun.rao@gmail.com	10

10 rows in set (0.00 sec)

```
mysql> select email, substring_index(email, '@') as extracted_email from employees;
ERROR 1582 (42000): Incorrect parameter count in the call to native function
'substring_index'
```

```
mysql> select email, substring_index(email, '@', 1) as extracted_email from
employees;
```

email	extracted_email
rahul.kumar@gmail.com	rahul.kumar
anita.sharma@yahoo.com	anita.sharma
suresh.verma@gmail.com	suresh.verma
priya.singh@outlook.com	priya.singh
rahul.joshi@gmail.com	rahul.joshi
kiran.reddy@gmail.com	kiran.reddy
deepika.naidu@yahoo.com	deepika.naidu

maresh.patel@gmail.com	maresh.patel
divya.gupta@gmail.com	divya.gupta
arjun.rao@gmail.com	arjun.rao

10 rows in set (0.00 sec)

```
mysql> select email, substring_index(email,'@',-1) as extracted_email from employees;
```

email	extracted_email
rahul.kumar@gmail.com	gmail.com
anita.sharma@yahoo.com	yahoo.com
suresh.verma@gmail.com	gmail.com
priya.singh@outlook.com	outlook.com
rahul.joshi@gmail.com	gmail.com
kiran.reddy@gmail.com	gmail.com
deepika.naidu@yahoo.com	yahoo.com
maresh.patel@gmail.com	gmail.com
divya.gupta@gmail.com	gmail.com
arjun.rao@gmail.com	gmail.com

10 rows in set (0.00 sec)

```
mysql> select * from employees where last_name like '%a';
```

emp_id	first_name	last_name	email	phone_number	department	city	salary
102	Anita	Sharma	anita.sharma@yahoo.com	912-345-6789	HR	Bangalore	38000.00
103	Suresh	Verma	suresh.verma@gmail.com	998-123-4567	Sales	Chennai	52000.00
109	Divya	Gupta	divya.gupta@gmail.com	955-777-8888	Finance	Delhi	49000.00

3 rows in set (0.03 sec)

```
mysql> select upper(concat(first_name,' ',last_name)) as upper_full_name from employees where department = 'IT';
```

upper_full_name
RAHUL KUMAR
KIRAN REDDY

2 rows in set (0.02 sec)

```
mysql> select first_name, count(first_name) as count group by first_name having
count > 1;
```

ERROR 1054 (42S22): Unknown column 'first_name' in 'field list'

```
mysql> select first_name, count(first_name) as count from employees group by
first_name having count > 1;
```

```
+-----+-----+
| first_name | count |
+-----+-----+
| Rahul      |      2 |
+-----+-----+
1 row in set (0.05 sec)
```

```
mysql> select phone_number, replace(phone_number, '-', '') as clean_number from
employees;
```

```
+-----+-----+
| phone_number | clean_number |
+-----+-----+
| 987-654-3210 | 9876543210   |
| 912-345-6789 | 9123456789   |
| 998-123-4567 | 9981234567   |
| 900-456-7890 | 9004567890   |
| 955-888-1111 | 9558881111   |
| 901-222-3333 | 9012223333   |
| 944-111-2222 | 9441112222   |
| 933-444-5555 | 9334445555   |
| 955-777-8888 | 9557778888   |
| 988-999-7777 | 9889997777   |
+-----+-----+
10 rows in set (0.00 sec)
```

```
mysql> select * from employees where first_name like '%i%';
```

```
+-----+-----+-----+-----+-----+-----+
| emp_id | first_name | last_name | email | phone_number |
+-----+-----+-----+-----+-----+-----+
| 102 | Anita | Sharma | anita.sharma@yahoo.com | 912-345-6789 | HR
| 104 | Priya | Singh | priya.singh@outlook.com | 900-456-7890 |
Finance | Pune | 47000.00 |
| 106 | Kiran | Reddy | kiran.reddy@gmail.com | 901-222-3333 | IT
| 107 | Deepika | Naidu | deepika.naidu@yahoo.com | 944-111-2222 | HR
| 109 | Divya | Gupta | divya.gupta@gmail.com | 955-777-8888 |
Finance | Delhi | 49000.00 |
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
| department | city | salary |
+-----+-----+-----+-----+-----+-----+
| 102 | Bangalore | 38000.00 |
| 104 | Pune | 47000.00 |
| 106 | Hyderabad | 56000.00 |
| 107 | Vizag | 39000.00 |
| 109 | Delhi | 49000.00 |
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
| department | city | salary |
+-----+-----+-----+-----+-----+-----+
```

5 rows in set (0.00 sec)

```
mysql> select * from employees order by length(first_name);
```

emp_id	first_name	last_name	email	phone_number	
department	city	salary			
101	Rahul	Kumar	rahul.kumar@gmail.com	987-654-3210	IT
	Hyderabad	45000.00			
102	Anita	Sharma	anita.sharma@yahoo.com	912-345-6789	HR
	Bangalore	38000.00			
104	Priya	Singh	priya.singh@outlook.com	900-456-7890	
Finance	Pune	47000.00			
105	Rahul	Joshi	rahul.joshi@gmail.com	955-888-1111	
Testing	Delhi	41000.00			
106	Kiran	Reddy	kiran.reddy@gmail.com	901-222-3333	IT
	Hyderabad	56000.00			
109	Divya	Gupta	divya.gupta@gmail.com	955-777-8888	
Finance	Delhi	49000.00			
110	Arjun	Rao	arjun.rao@gmail.com	988-999-7777	
Developer	Hyderabad	68000.00			
103	Suresh	Verma	suresh.verma@gmail.com	998-123-4567	Sales
	Chennai	52000.00			
108	Mahesh	Patel	mahesh.patel@gmail.com	933-444-5555	Sales
	Mumbai	62000.00			
107	Deepika	Naidu	deepika.naidu@yahoo.com	944-111-2222	HR
	Vizag	39000.00			

10 rows in set (0.00 sec)

```
mysql> select * from employees where length(city) > 6;
```

emp_id	first_name	last_name	email	phone_number	
department	city	salary			
101	Rahul	Kumar	rahul.kumar@gmail.com	987-654-3210	IT
	Hyderabad	45000.00			
102	Anita	Sharma	anita.sharma@yahoo.com	912-345-6789	HR
	Bangalore	38000.00			
103	Suresh	Verma	suresh.verma@gmail.com	998-123-4567	Sales
	Chennai	52000.00			
106	Kiran	Reddy	kiran.reddy@gmail.com	901-222-3333	IT
	Hyderabad	56000.00			
110	Arjun	Rao	arjun.rao@gmail.com	988-999-7777	
Developer	Hyderabad	68000.00			

```
+-----+-----+-----+-----+-----+-----+
-----+-----+-----+
5 rows in set (0.00 sec)
```

```
mysql> notee
```